

KITEWRIGHT RANGE · DESIGN DECISIONS REGISTER

Kitewright Range design decisions register

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DOCUMENT CONTROL

Document ID	KWR-DEC-001	Version	0.1
Status	Draft	Date	2026-10-03
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Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/kitewright-range		
Source	docs/06-design-decisions.md @ uncommitted		

REVISION HISTORY

Version	Date	Author	Change
0.1	2026-10-03	Amish Chadha	Register opened at TRL 3; decisions made under the 2026-10-03 pre-approval; four requirement decisions proposed, awaiting Amish

Kitewright Range design decisions register

Every design decision still to be made, and every decision made, in one place. Each decision is argued in full in its decision record under `docs/decisions/`, and the open ones in `docs/REVIEW.md` (TRL 3, Decisions for Amish); this register is the index Amish works from. The build plan (`docs/05-build-plan.md`) describes the design as it stands and does not list open decisions.

■ Open decisions

All four are about requirements the design does not meet or meets too thinly. They are **Proposed, awaiting Amish**; the design and build plan stay as they are until he decides. Figures are from KWR-CAL-001 v0.2, Table 5.

#	Decision needed	Options	Recommendation	Affects in the build	Source
O1	R1 hover thrust margin: 31.8 % on unmeasured maker-class thrust	A: keep 20 in propellers on 5212-class motors (31.8 %, no cost or mass change). B: 22 in propellers on 5215-class motors (53.3 %, +USD 160, +0.40 kg, endurance -1.7 min; booms 50 mm longer and 20 mm further out). C: keep A and change only if AltiRig measures under the target	B , the conservative choice: hover authority at 5,000 m is a safety margin	Lift motors, propellers, booms, pylon position	KWR-CAL-001 C, H
O2	R2 endurance: 31.0 min	A: two 6S3P packs of 5.0 Ah cells (35.6 min, +USD 120, no mass change). B: two 6S4P packs (38.9 min, +USD 190, +0.88 kg; hover margin falls to 21.7 %). C: moulded carbon airframe with two 6S4P packs (43.3 min, +USD 1,390, +0.19 kg; margin 29.4 %)	A for the first prototype; C as the second-prototype path	ColdCell cell choice only	KWR-CAL-001 D, H
O3	R3 survey range: 40.9 km	A: 5.0 Ah cells as O2-A (46.8 km, +USD 120). B: with O1-B as well (45.0 km, +USD 280, +0.40 kg). C: keep as is (40.9 km)	A , taken together with O2-A (and B if O1-B is chosen)	ColdCell cell choice only	KWR-CAL-001 D, H
O4	R7 take-off mass: 10.58 kg	A: keep the foam, glass and lite-ply airframe (10.58 kg). B: moulded carbon wing, tail and fuselage (9.89 kg, +USD 1,200, endurance +4.1 min; USD 5,145 in all, USD 145 over the value-engineering target). C: fly one pack (9.18 kg; endurance 15.0 min)	A for the first prototype, weighed at TRL 4; no option inside the concept reaches the target without losing R2	None for A	KWR-CAL-001 A, H

■ To confirm when parts are bought

#	What to confirm	Why it matters	Source
1	Lift motor and propeller thrust at 0.736 kg/m ³ , measured in AltiRig	R1 rests on maker-class figures scaled by density	KWR-CAL-001 C
2	Cold ratings of the ESCs, servos, cruise motor ESC and pitot heater to -20 °C, or a cold soak test	R5 is met only by specification	KWR-REQ-001
3	The joiner slides into both spar bores without play	The wing joint depends on it	KWR-DDR-002, C6
4	The cruise motor's bolt circle and shaft length	They set the nose cone inserts and the propeller position	KWR-DDR-002, C5
5	Kitewright Core avionics envelope (180 x 110 x 60 mm) and mass (0.78 kg with the mast)	They set the front bay and the balance	Cross-repo assumption
6	ColdCell pack size (138 x 75 x 82 mm), mass (1.40 kg) and XT90 lead	They set the battery bays and the balance	Cross-repo assumption
7	Payload mount plate (160 x 80 x 12 mm) and its four M4 holes	They set the floor inserts	Cross-repo assumption
8	Transport cases with at least 1,250 mm inside	The wing panel is 1,205 mm with its root fittings	KWR-CAL-001 F

■ Value engineering

Value-engineering target: USD 5,000 (a hypothetical control target, not a limit). Estimated cost of the constructable design: USD 3,945 (USD 1,055 under the target). Main cost drivers and savings worth trying:

- The largest lines are the Kitewright Core avionics share (USD 900, assumed), the two ColdCell packs (USD 580), the four lift motors (USD 480), the two transport cases (USD 360) and the four lift ESCs (USD 180).
- Savings worth trying: soft padded bags in place of hard cases for vehicle transport (about USD 200); four-in-one ESC boards are lighter but would move power wiring into the fuselage, so they are not recommended.
- The open decisions add at most USD 1,390 (O2-C); the recommended set (O1-B with O2-A) adds USD 280 and keeps the design under the target.

■ Decisions made

Date	Decision	Decided by	Record
2026-10-03	TRL 2 review decisions D1 to D10: PX4; 2.5 m two-piece wing; one 20 in propeller set; winch on Lift; nose tractor and tail boom; two ColdCell 6S3P packs; foam, ply and printed build; Sikkim State Disaster Management Authority as first co-design candidate (not agreed); staged powered tests; conservative failsafes	Amish: "start with the first 14 repos from the list of 29 projects. I pre-approve the batch runs along with any recommendations you come up with. I also accept any cost overruns or variations from the assumed scope cost." and "Proceed with the remaining 15 scaffolds"	KWR-DDR-001
2026-10-03	Design for construction C1 to C13: tail boom, pylons, boom spacing, motors on the centre of gravity, nose tractor, joiner and wing bolts, fuselage bays, landing legs, wing conduits, payload mount, tail joint, hatch, pitot	Amish, same pre-approval	KWR-DDR-002